



KOM4531 ADVANCED INDUSTRIAL AUTOMATION SYSTEMS APPLICATIONS

IMPORTANT CONCEPTS



- What is MAC (Media Access Control) ?
- What is IP (Internet Protocol) ?
- What is DNS (Domain Name Service) ?
- What is ARP (Address Resolution Protocol) ?

IMPORTANT CONCEPTS

MAC



It is not a node that has a LAN address but instead a node's adapter that has a LAN address. A LAN address is also variously called a physical address an Ethernet address, or a MAC (media access control) address.

For most LANs (including Ethernet and token-passing LANs), the LAN address is six-bytes long, giving 248 possible LAN addresses. These six-byte addresses are typically expressed in hexadecimal notation, with each byte of the address expressed by a pair of hexadecimal numbers.

An adapter's LAN address is permanent when an adapter is manufactured, a LAN address is burned into the adapter's ROM. So a LAN address can not be changed.

IMPORTANT CONCEPTS IP



- You can use the same IP addresses on two distinct local network.
- For IP assignment, You need to assign the IP address, Subnet Mask and Gateway.
 - IP address is a 32 bit long number and it is divided with dots. Examples: 192.168.0.5 or 152.236.87.49. In a network, all devices should have different IP addresses.
 - Subnet Mask handles the neighborhood size. For a subnet mask of 255.255.255.0 , there can be 254 device in the network, and their IP's first three part should be same and the last part should be different.
 - Gateway is needed if the transmission package is going out of that network. By default the first IP of the neighborhood is gateway address. For example 192.168.0.1 or 192.168.1.1.

IMPORTANT CONCEPTS

DNS



- One identifier for a host is its hostname. Hostnames such as cnn.com, www.yahoo.com, are mnemonic and are therefore appreciated by humans. However, hostnames provide little, if any, information about the location within the Internet of the host.
- For these reasons, hosts are also identified by so-called IP addresses. An IP address consists of four bytes and has a rigid hierarchical structure. An IP address looks like 121.7.106.83, where each period separates one of the bytes expressed in decimal notation from 0 to 127.

IMPORTANT CONCEPTS

ARP



- ARP resolves an IP address to a LAN address. In many ways it is analogous to DNS which resolves hostnames to IP addresses. One important difference between the two resolvers is that DNS resolves hostnames for hosts anywhere in the Internet, whereas ARP only resolves IP addresses for nodes on the same LAN.
- The ARP module in each node has a table in its RAM called an ARP table. This table contains the mappings of IP addresses to LAN addresses. For each address mapping the table also contains a time-to-live (TTL) entry, which indicates when the entry will be deleted.
- The table does not necessarily contain an entry for every node on the LAN; some nodes may have had entries that expired over time, whereas other nodes may have never been entered into the table.

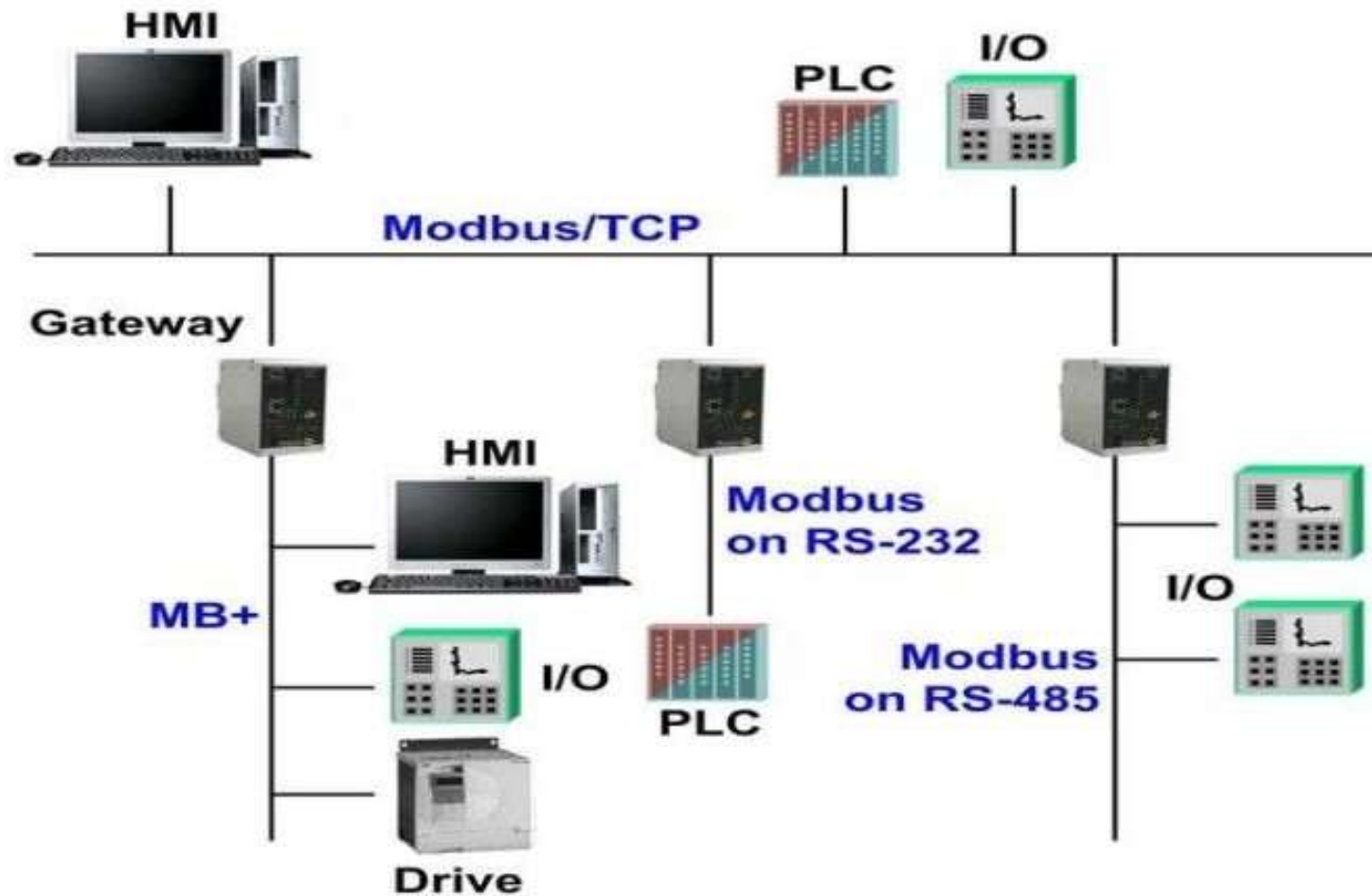
IP address	LAN address	TTL
222.222.222.221	88-B2-2F-54-1A-0F	13:45:00
222.222.222.223	5C-66-AB-90-75-B1	13:52:00

IMPORTANT CONCEPTS



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- What is IP (Internet Protocol) ?
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- What is ARP (Address Resolution Protocol) ?

MODBUS PROTOCOL



MODBUS PROTOCOL



MODBUS Master/Slave Structure

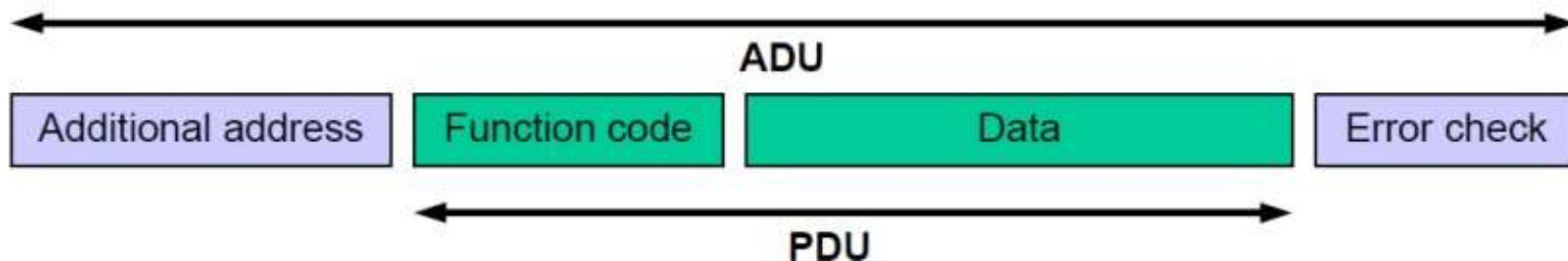
- ▶ In Modbus communication, the querying process is performed by the device called Master. This device can also be named as a Client in the Modbus TCP protocol.
- ▶ The request from the master or client is answered by the device called Slave or Server.
- ▶ Generally, in Modbus communication structures, a Master queries one or more Slave devices.

MODBUS PROTOCOL



MODBUS RTU Communication

- ▶ The Modbus application protocol creates a request format initiated by the master device and tells the application data unit and what kind of operation it will do to which Slave device.



MODBUS PROTOCOL



- ▶ The function code field of the Modbus data unit is coded by one byte, and the valid value range is decimal numbers from 1 to 255 (125 to 255 reserved and used for error conditions).
- ▶ As a function code, 0 is not a valid value. These function codes determine what kind of operation the slave devices will do.

MODBUS PROTOCOL



- ▶ If there is no error in the application data unit requested by the master device, the Slave device performs the requested operation and responds to the request of the Master device.
- ▶ However, if an error occurs, the Slave device sends an error message to the Master device, allowing the operations to continue.

MODBUS PROTOCOL



- ▶ When the slave device responds to the master device, it uses the function code field either to indicate a normal response or to report that some kind of error has occurred.
- ▶ In the case of a normal response, the Slave device uses the original requesting function code.

MODBUS PROTOCOL



- ▶ The size of the Modbus protocol data unit is limited by size constants inherited from the first development of the protocol over the serial line.
- ▶ RS232 / RS485 ADU = 253 bytes + Address (1 byte) + CRC (2 bytes) = 256 bytes
- ▶ Modbus TCP ADU = 253 bytes + MBAP (7 bytes) = 260 bytes

MODBUS PROTOCOL



► Modbus Register Structure

Coil/Register Numbers	Data Addresses	Type	Table Name
1-9999	0000 to 270E	Read-Write	Discrete Output Coils
10001-19999	0000 to 270E	Read-Only	Discrete Input Contacts
30001-39999	0000 to 270E	Read-Only	Analog Input Registers
40001-49999	0000 to 270E	Read-Write	Analog Output Holding Registers

MODBUS PROTOCOL

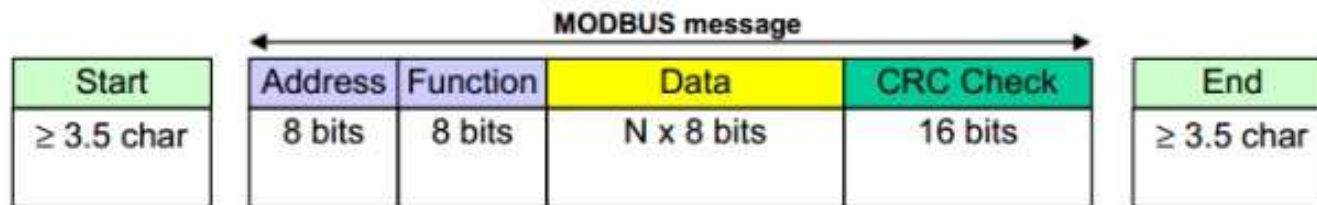
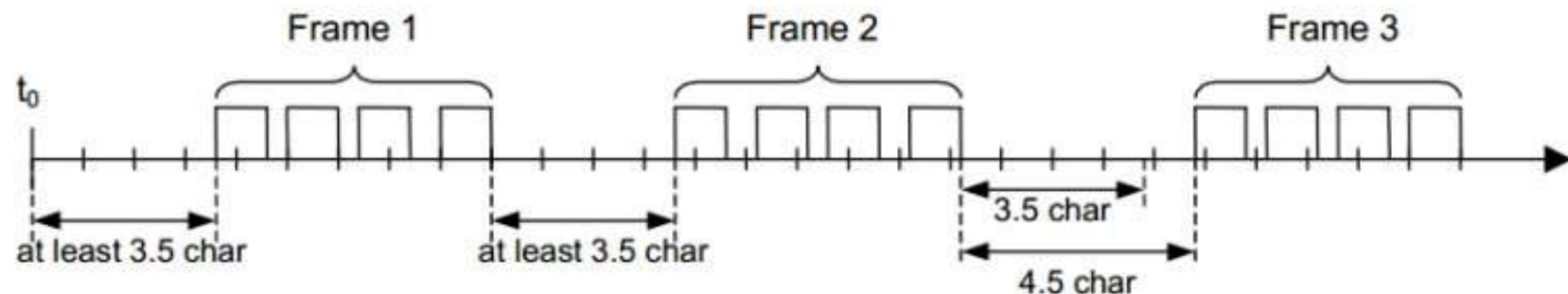


- ▶ Characters or byte data are sent serially from left to right, from most significant bit to least significant bit.
- ▶ The first bit sent, respectively, is called the start bit (start bit), then 8-bit data bits, followed by a parity bit and stop bit (if any) are sent.

MODBUS PROTOCOL



- ▶ The Modbus message is placed in a data area whose beginning and end are known by the device that will transmit the message.
- ▶ In RTU mode, data fields are separated by a waiting time of at least 3.5 characters.



MODBUS PROTOCOL



MODBUS TCP

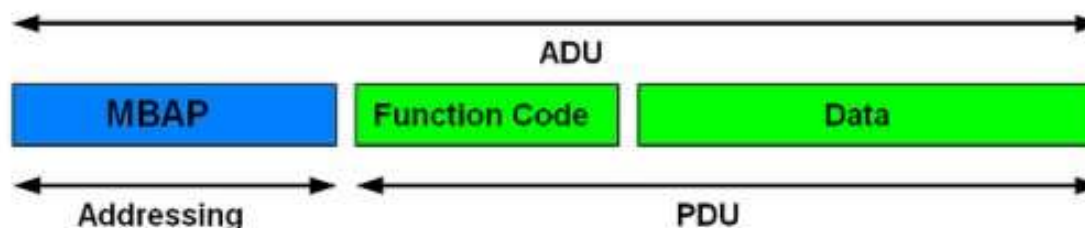
General MODBUS Frame



MODBUS/RTU Serial Frame



MODBUS/TCP Frame



MODBUS PROTOCOL



► Function Code List

Function code	Related area	Operation
1	Coils	Read up to 2000 contiguous memory bits
2	Discrete Inputs	Read up to 2000 contiguous input bits
3	Holding Registers	Read up to 125 contiguous memory words
4	Input Registers	Read up to 125 contiguous input words
5	Single Coil	Write one memory bit
6	Single Register	Write one memory word
15	Coils	Write up to 1968 contiguous memory bits
16	Holding Registers	Write up to 123 contiguous memory words
23	Holding Registers	Read up 125 and write up 121 memory words

MODBUS PROTOCOL



MODBUS Function Codes (FC1)

Request

Function code	1 Byte	0x01
Starting Address	2 Bytes	0x0000 to 0xFFFF
Quantity of coils	2 Bytes	1 to 2000 (0x7D0)

Response

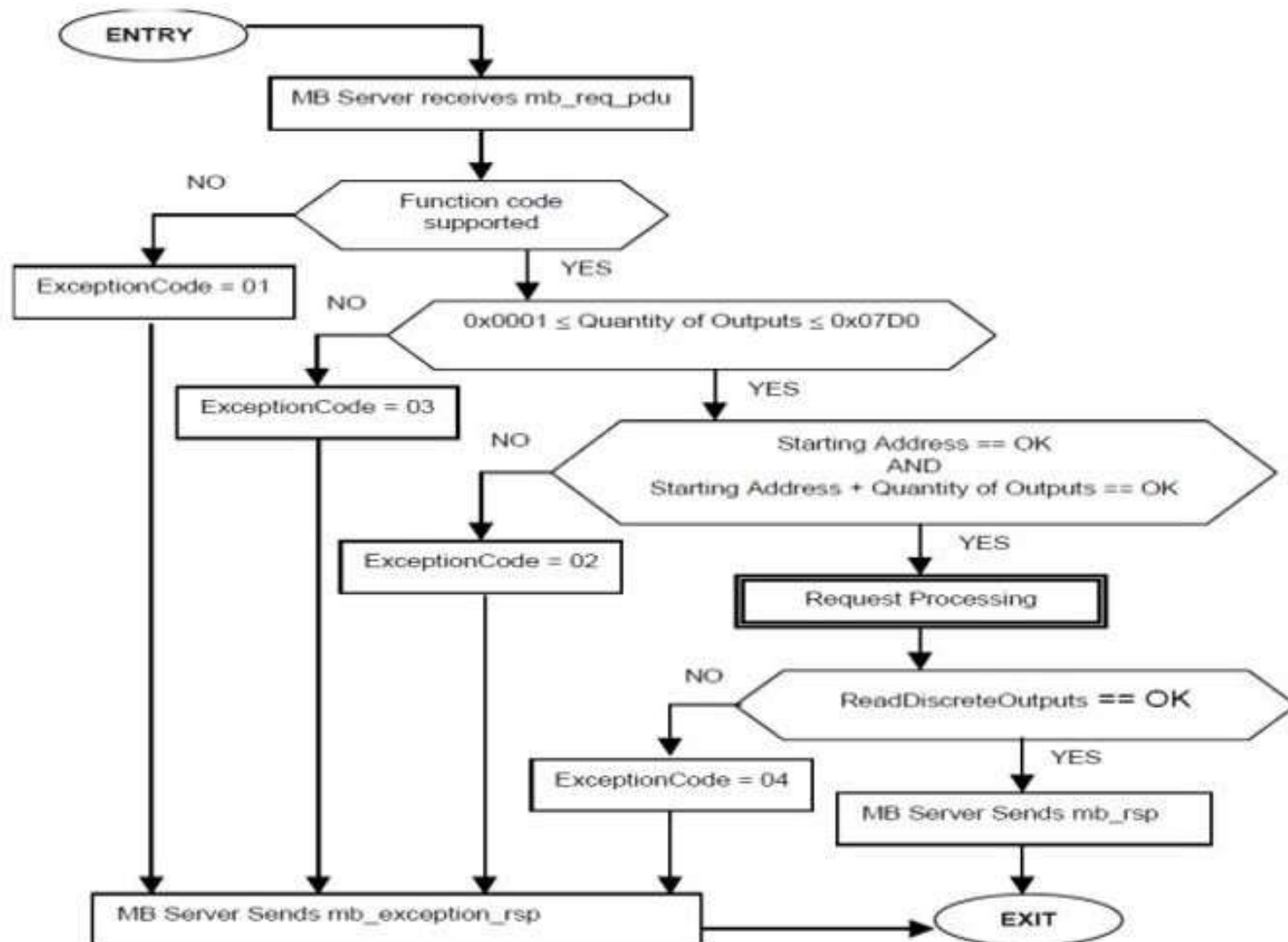
Function code	1 Byte	0x01
Byte count	1 Byte	N*
Coil Status	n Byte	n = N or N+1

*N = Quantity of Outputs / 8, if the remainder is different of 0 $\Rightarrow$ N = N+1

Error

Function code	1 Byte	Function code + 0x80
Exception code	1 Byte	01 or 02 or 03 or 04

MODBUS PROTOCOL



MODBUS PROTOCOL



MODBUS Function Codes (FC3)

Request

Function code	1 Byte	0x03
Starting Address	2 Bytes	0x0000 to 0xFFFF
Quantity of Registers	2 Bytes	1 to 125 (0x7D)

Response

Function code	1 Byte	0x03
Byte count	1 Byte	2 x N*
Register value	N* x 2 Bytes	

*N = Quantity of Registers

Error

Error code	1 Byte	0x83
Exception code	1 Byte	01 or 02 or 03 or 04

QUESTIONS ?